

1 PURPOSE

Define a general finite-element plate solver for slab behavior.

PRIMARY OUTPUTS

Displacements	nodal w , θ_x , θ_y or selected plate DOFs
Reactions	support reactions at constrained DOFs and support regions
Support actions	resultants such as V , M_x , M_y over finite supports
Fields	optional moments, shears, stresses, deflections, contours
Diagnostics	convergence, equilibrium, mesh quality, support resolution

2 CONSUMERS

The solver is not a punching-shear solver. Punching shear is the first consumer of support actions.

DOWNSTREAM USES

Punching shear	consume V_u, M_{ux}, M_{uy} and run critical-section J_c check
Slab transfer	integrate column/wall reactions and support eccentricity
Deflection	report service displacement fields and envelopes
Validation	compare support reactions, deflections, and moment fields to references
Design workflows	future strip forces, envelopes, and reinforcement design inputs

3 REFERENCE TARGET

ETABS/SAFE-style slab analysis is the validation target. The goal is not to copy private implementation details, but to match the public engineering behavior closely enough that benchmark models produce the same reactions, support moments, deflections, and downstream punching demands within documented tolerances.

TARGET BEHAVIOR

Shell/plate model	slab represented by meshed plate/shell finite elements
Object supports	point, line, wall, column, spring, and area/patch supports
Rigid zones	finite column/support regions avoid point-singularity behavior
Load cases	solve individual cases and factored combinations explicitly
Postprocessing	recover support actions, fields, envelopes, and code-check inputs

Benchmark success means the same physical model, loading, supports, and mesh intent give matching results against ETABS/SAFE, not merely internal equilibrium.

4 DEVELOPMENT REALITY

A useful solver kernel can be built before ETABS/SAFE automation exists, but it cannot be accepted as an ETABS/SAFE clone until benchmark comparison is running.

LOCAL WORK WITHOUT CSI

Element unit tests	stiffness symmetry, rigid-body modes, patch tests
Load-vector tests	applied load sums and consistent nodal loading
Constraint tests	point, line, spring, patch, and settlement mapping
Equilibrium tests	recovered reactions balance applied load
Closed-form checks	simple square/strip plate deflection benchmarks
Symmetry checks	mirrored models produce mirrored reactions and moments

REQUIRES ETABS/SAFE

Benchmark parity	reactions, moments, deflections, and field results vs CSI
Support tuning	column rigid zones, wall supports, springs, patch behavior
Meshing tolerance	how fine the mesh must be for CSI-like support actions
Punching demands	V_u, M_{ux}, M_{uy} matching SAFE-style punching inputs
Production acceptance	release only after fixture set passes tolerances

5 TUNING PHILOSOPHY

Treat ETABS/SAFE as the reference oracle for practical two-way slab behavior. Tune only documented modeling choices, not arbitrary correction factors.

TUNABLE MODELING CHOICES

Element formulation	DKT vs Mindlin/Reissner or later mixed element
Support representation	point, line, spring, rigid patch, elastic patch
Rigid zone extent	finite column/support footprint behavior
Mesh strategy	object-based meshing, local refinement, element shape limits
Result smoothing	raw element fields vs averaged display/design fields
Load application	area, line, point, and combination handling

Do not tune by applying hidden multipliers to results. If a benchmark does not match, first identify whether the cause is element formulation, support model, mesh, load case definition, units, or result recovery.

6 MODEL INPUTS

Geometry	slab polygon, openings, levels, thickness regions
Material	E_c , ν , density, cracking modifiers if used
Loads	area, line, point, thermal, settlement, load cases and combinations
Supports	point pins, point springs, line/wall supports, finite patches, rotational springs
Mesh controls	target edge length, local refinement, support footprint resolution

7 COORDINATE SYSTEM

$$x, y = \text{slab plan axes}, \quad z = \text{vertical} \quad (1)$$

Use one explicit sign convention throughout the solver.

PROPOSED CONVENTION

w	positive downward vertical displacement
q	positive downward pressure
V	positive downward force transferred from slab to support
M_x, M_y	right-hand-rule support moments about global axes, with signs documented

8 ELEMENT FORMULATION

Open decision: choose the production plate element.

CANDIDATES

DKT triangle	thin-plate bending, three DOFs per node, good for Kirchhoff plates
Mindlin/Reissner	includes transverse shear, needs locking control for thin slabs
Mixed formulation	possible future route for shear forces and thick/slender robustness

For the first solver contract, only require that each element provides:

$$\mathbf{K}_e, \quad \mathbf{f}_e, \quad \text{element recovery operators} \quad (2)$$

9 LOAD VECTOR

Convert loads into consistent element or nodal load vectors.

$$\mathbf{f}_e = \int_{A_e} \mathbf{N}^T q dA \quad (3)$$

For a simple triangular element with linear vertical displacement interpolation:

$$F_{w,i}^{(e)} = \frac{qA_e}{3} \quad (4)$$

Open item: define how rotational load terms are handled for the chosen plate element.

10 SUPPORT TYPES

The solver must support multiple boundary-condition families.

SUPPORT CATALOG

Point pin	constrain w at a node; rotations free
Point fixed	constrain w, θ_x, θ_y at a node
Point spring	add translational and/or rotational spring stiffness
Line/wall support	constrain or spring nodes along a wall/polyline
Finite patch	support over a column, bearing plate, wall end, or reaction area
Rigid patch	finite patch with master-slave rigid vertical kinematics
Elastic foundation	distributed Winkler/Pasternak style support over an area
Settlement	prescribed support displacement at constrained or spring-supported DOFs

11 POINT AND LINE SUPPORTS

Pinned point support:

$$w_i = 0 \quad (5)$$

Rotationally fixed point support:

$$w_i = 0, \quad \theta_{x,i} = 0, \quad \theta_{y,i} = 0 \quad (6)$$

Spring support:

$$\mathbf{K} \leftarrow \mathbf{K} + \mathbf{K}_s \quad (7)$$

Line supports are modeled by applying compatible point constraints or springs to mesh nodes on the support line, or by a distributed support element.

12 FINITE SUPPORT PATCHES

Finite patches are required when the support footprint matters.

EXAMPLES

Column footprint	rectangular or polygonal area around column
Bearing plate	finite support under equipment or transfer load
Wall strip	finite-width wall support, not just a line
Odd geometry	clipped or rotated reaction areas

Mesh requirements:

Master node	at patch centroid when rigid patch is used
Slave nodes	enough mesh nodes inside patch to avoid point-support behavior
Refinement	target edge tied to support size, not only slab span
Diagnostics	slave count, master offset, patch area, clipped footprint

13 RIGID PATCH KINEMATICS

For rigid vertical patch behavior:

$$w_s = w_c + \theta_y \Delta x - \theta_x \Delta y \quad (8)$$

Transformation form:

$$\mathbf{u} = \mathbf{T}\mathbf{q}, \quad \mathbf{K}_r = \mathbf{T}^T \mathbf{K} \mathbf{T}, \quad \mathbf{F}_r = \mathbf{T}^T \mathbf{F} \quad (9)$$

Open item: decide when a rigid patch is mandatory versus optional.

14 ASSEMBLY AND SOLVE

Assemble:

$$\mathbf{K} = \sum_e \mathbf{A}_e^T \mathbf{K}_e \mathbf{A}_e, \quad \mathbf{F} = \sum_e \mathbf{A}_e^T \mathbf{f}_e \quad (10)$$

Apply constraints and solve:

$$\mathbf{K}_r \mathbf{q} = \mathbf{F}_r \quad (11)$$

Then expand to full DOFs:

$$\mathbf{u} = \mathbf{T} \mathbf{q} \quad (12)$$

15 REACTION RECOVERY

At a constrained DOF:

$$R_i = (\mathbf{K} \mathbf{u})_i - F_i \quad (13)$$

Demand on the support is the opposite sign:

$$P_i = F_i - (\mathbf{K} \mathbf{u})_i \quad (14)$$

For springs, the support demand can also be recovered from spring deformation:

$$\mathbf{P}_s = \mathbf{K}_s (\mathbf{u} - \mathbf{u}_0) \quad (15)$$

16 SUPPORT RESULTANTS

For any finite support region with vertical demand samples P_i :

$$V = \sum_i P_i \quad (16)$$

$$M_x = \sum_i (y_i - y_c) P_i \quad (17)$$

$$M_y = - \sum_i (x_i - x_c) P_i \quad (18)$$

Equivalent eccentricity form:

$$x_R = \frac{\sum_i x_i P_i}{V}, \quad y_R = \frac{\sum_i y_i P_i}{V} \quad (19)$$

$$e_x = x_R - x_c, \quad e_y = y_R - y_c \quad (20)$$

$$M_x = V e_y, \quad M_y = -V e_x \quad (21)$$

These resultants are general. Punching shear is one workflow that consumes them.

17 LOCAL AXES

For a support patch rotated by θ :

$$M_1 = M_x \cos \theta + M_y \sin \theta \quad (22)$$

$$M_2 = -M_x \sin \theta + M_y \cos \theta \quad (23)$$

Open item: define local-axis convention once and reuse it for columns, walls, loads, and punching checks.

18 FIELD RECOVERY

Open recovery targets:

Deflection field	nodal and interpolated w
Rotation field	θ_x, θ_y
Plate moments	M_x, M_y, M_{xy} from element recovery
Plate shears	Q_x, Q_y if supported by element formulation
Contours	smoothed or element-native display values

19 DIAGNOSTICS

Equilibrium	recovered reactions balance applied loads
Convergence	solver residual and iteration count
Mesh quality	element shape, fallback tier, local refinement
Support resolution	patch slave count, master offset, footprint coverage
Conditioning	warnings for weak restraints or near-mechanisms

20 ETABS/SAFE VALIDATION

Validation should progress from simple closed-form checks to CSI comparison fixtures.

COMPARISON OUTPUTS

Global equilibrium	total vertical reactions vs total applied load
Support reactions	per-column and per-wall vertical reactions
Support moments	column/support M_x, M_y and local-axis moments
Deflections	selected nodal and maximum slab deflections
Field recovery	plate moment contours where element assumptions match
Punching demands	V_u, M_{ux}, M_{uy} passed into the J_c check

FIXTURE SET

Single-panel slab	closed-form and CSI comparison
Column grid	interior, edge, and corner support actions
Wall-supported slab	line support reactions and moments
Rotated columns	local/global moment transformation
Openings/irregular slab	mesh clipping and support-action stability
Patch sensitivity	point support vs finite patch vs rigid zone behavior

Each comparison fixture should store the exported CSI model setup, exported result tables, solver input JSON, and tolerance notes.

ACCEPTANCE GATE

Local solver gate	closed-form, symmetry, and equilibrium tests pass
CSI parity gate	curated ETABS/SAFE fixtures pass stated tolerances
Stability gate	unstable support/action diagnostics block downstream design checks
Regression gate	every fixture remains reproducible from stored inputs and exports

21 CHECK ORDER

- 1 Build plate geometry, materials, loads, supports, and mesh controls
- 2 Mesh slab with refinement near supports, openings, and result regions
- 3 Assemble element stiffness and load vectors
- 4 Apply point, line, spring, patch, settlement, and rigid constraints
- 5 Solve reduced plate system
- 6 Expand full displacement vector
- 7 Recover reactions, support resultants, and field quantities
- 8 Run diagnostics and mark each recovered result stable/degraded/unstable
- 9 Compare benchmark fixtures against ETABS/SAFE exports before production use
- 10 Pass accepted recovered artifacts to downstream workflows

22 REVIEW QUESTIONS

- Q1 Which element formulation should be production v1: DKT or Mindlin?
- Q2 What support types are required for v1 beyond point, line, and finite patch?
- Q3 When should a finite patch use rigid kinematics versus elastic distribution?
- Q4 What mesh quality and support-resolution gates make a result unstable?
- Q5 Should support resultants always be recovered by force integration, spring reaction, or both?
- Q6 Which ETABS/SAFE settings must be mirrored for benchmark parity?
- Q7 What tolerances are acceptable for reactions, moments, deflections, and field outputs?
- Q8 What result artifacts should be persisted for model-space visualization?